

Nicklas Hansen

hello@nicklashansen.com | nicklashansen.com | [nicklashansen](https://github.com/nicklashansen) | [ncklas](https://www.linkedin.com/in/ncklas) | [San Diego, CA](#)

Education

University of California, San Diego

PhD student, Computer Science and Engineering, GPA: 3.85/4.0
 · Advised by Xiaolong Wang and Hao Su. NVIDIA Graduate Fellow.

San Diego, CA, USA

Sep 2021 - June 2026

University of California, Berkeley

Visiting Student, GPA: 4.0/4.0
 · Spar Nord Fonden's FinTech scholarship recipient. SCET's Collider Cup finalist.

Berkeley, CA, USA

Spring 2020

Technical University of Denmark

MS Mathematical Modeling & Computation, GPA: 11.2/12.0
 · Special topics in machine learning. Advised by Ole Winther.

Kongens Lyngby, Denmark

Feb 2019 - Feb 2021

Technical University of Denmark

BS Software Technology, GPA: 8.2/12.0
 · **Nanyang Technological University, Singapore** - semester abroad, Fall 2017.

Kongens Lyngby, Denmark

Sep 2015 - Dec 2018

Publications & Preprints (37)

SNAP: One-Shot Dexterous Manipulation via Functional Grasping Policies

Paper under review
 Alexiy Buynitsky, **Nicklas Hansen**, Carmelo Sferrazza, Jianglong Ye, Xiaolong Wang
 preprint available soon

Under review

2026

Causal World-Action Models for Failure-Aware Robot Policies

Paper under review
 Quanquan Peng, Yutong Liang, Rui Yan, **Nicklas Hansen**, Xiaolong Wang
 preprint available soon

Under review

2026

Simulation Priors for Rapid Adaptation in Model-Based RL for Autonomous Racing

Paper under review
 Adrian Remonda, Jiajun Xi, **Nicklas Hansen**, Marcus Greiff, Thomas Lew, John Talbot,
 John Subosits, Xiaolong Wang
 preprint available soon

Under review

2026

Generating Robot Hands from Human Demonstrations

Paper under review
 Yi Sha, **Nicklas Hansen**, Xueqian Bai, Carmelo Sferrazza, Michael T. Tolley, Xiaolong Wang
 preprint available soon

Under review

2026

Hallucination in World Models is Predictable and Preventable

Paper under review
Nicklas Hansen, Xiaolong Wang
 preprint available soon

Under review

2026

HiGym: Learning to Act in High-Dimensional Spaces

Paper under review
 Harshit Dhankhar, Xiaolong Wang, **Nicklas Hansen**
 preprint available soon

Under review

2026

Bayesian Curriculum Learning over Latent Geometry in Large Language Models

Paper under review

Darrien McKenzie, **Nicklas Hansen**, Xiaolong Wang
preprint available soon

Under review
2026

Catch It All!: Generalizable Dynamic Catching with a Dexterous Hand

Paper under review

Arth Shukla*, Kehlani Fay*, Huaze Liu^, Darin Djapri^, Alexiy Buynitsky, **Nicklas Hansen**
preprint available soon

Under review
2026

Learning Reasoning-Guided Robot Policies via Task-Aligned Simulated Futures

Paper under review

Hanan Gani, Tejal Kulkarni, Madhoolika Venkata Chodavarapu, **Nicklas Hansen**, Manmohan Chandraker
preprint available soon

Under review
2026

Abstracting Task-Centric World Models from Visual Foundations

Paper under review

Minghao Fu, Fan Feng, **Nicklas Hansen**, Biwei Huang
<https://arxiv.org/abs/2605.25620>

Preprint
2026

TIPS: Turn-Level Information-Potential Reward Shaping for Search-Augmented LLMs

International Conference on Learning Representations (ICLR)

Yutao Xie, Nathaniel Thomas, **Nicklas Hansen**, Yang Fu, Li Erran Li, Xiaolong Wang
<https://arxiv.org/abs/2603.22293>

2026

Learning Massively Multitask World Models for Continuous Control

International Conference on Learning Representations (ICLR)

Nicklas Hansen, Hao Su*, Xiaolong Wang*
<https://arxiv.org/abs/2511.19584>

2026

Learning to Design Soft Hands using Reward Models

International Conference on Robotics and Automation (ICRA)

Xueqian Bai, **Nicklas Hansen**, Adabhav Singh, Michael T. Tolley, Yan Duan, Pieter Abbeel, Xiaolong Wang, Sha Yi
<https://arxiv.org/abs/2510.17086>

2025

MyoChallenge 2024: A New Benchmark for Physiological Dexterity and Agility in Bionic Humans

Conference on Neural Information Processing Systems (NeurIPS)

Cheryl Wang, Chun Kwang Tan, [...], **Nicklas Hansen**, [...], Vikash Kumar, Vittorio Caggiano (26 authors)
<https://openreview.net/forum?id=1dSLbhErNv>

2025

Generalizable Robotic Insertion with World Models

Workshop on Out-of-Distribution Generalization in Robotics @ RSS

Nicklas Hansen, Iretiayo Akinola, Yijie Guo, Jie Xu, Bingjie Tang, Hao Su, Xiaolong Wang, Abhishek Gupta, Dieter Fox, Yashraj Narang
<https://openreview.net/pdf?id=DR3n6lqGKI>

2025

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

International Conference on Machine Learning (ICML)

Adrià López Escoriza, **Nicklas Hansen**, Stone Tao, Tongzhou Mu, Hao Su
<https://arxiv.org/abs/2503.01837>

2025

Policy Learning with Multi-Task World Models

International Conference on Learning Representations (ICLR)

Ignat Georgiev, Varun Giridhar, **Nicklas Hansen**, Animesh Garg
<https://arxiv.org/abs/2407.02466>

2025

Hierarchical World Models as Visual Whole-Body Humanoid Controllers International Conference on Learning Representations (ICLR) Nicklas Hansen , Jyothir S V, Vlad Sobal, Yann LeCun, Xiaolong Wang*, Hao Su* https://arxiv.org/abs/2405.18418	2025
A Simulation Benchmark for Autonomous Racing with Large-Scale Human Data Conference on Neural Information Processing Systems (NeurIPS) Adrian Remonda, Nicklas Hansen , Ayoub Raji, Nicola Musiu, Marko Bertogna, Eduardo E. Veas, Xiaolong Wang https://arxiv.org/abs/2407.16680	2024
A Recipe for Unbounded Data Augmentation in Visual Reinforcement Learning Reinforcement Learning Conference (RLC) Abdulaziz Almuzairee, Nicklas Hansen , Henrik I. Christensen https://arxiv.org/abs/2405.17416	2024
TD-MPC2: Scalable, Robust World Models for Continuous Control International Conference on Learning Representations (ICLR) Nicklas Hansen , Hao Su*, Xiaolong Wang* https://arxiv.org/abs/2310.16828	Spotlight 2024
Open X-Embodiment: Robotic Learning Datasets and RT-X Models International Conference on Robotics and Automation (ICRA) Open X-Embodiment Collaboration, [...], Nicklas Hansen , [...] (173 authors) https://arxiv.org/abs/2310.08864	Best Conference Paper 2024
MoDem-V2: Visuo-Motor World Models for Real-World Robot Learning International Conference on Robotics and Automation (ICRA) Patrick Lancaster, Nicklas Hansen , Aravind Rajeswaran, Vikash Kumar https://arxiv.org/abs/2309.14236	2024
Finetuning Offline World Models in the Real World Conference on Robot Learning (CoRL) Yunhai Feng*, Nicklas Hansen* , Ziyang Xiong*, Chandramouli Rajagopalan, Xiaolong Wang https://arxiv.org/abs/2310.16029	Oral 2023
Multi-Task Real Robot Learning with Generalizable Neural Feature Fields Conference on Robot Learning (CoRL) Yanjie Ze, Ge Yan, Yueh-Hua Wu, Annabella Macaluso, Yuying Ge, Jianglong Ye, Nicklas Hansen , Li Erran Li, Xiaolong Wang https://arxiv.org/abs/2308.16891	Oral 2023
On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline International Conference on Machine Learning (ICML) Nicklas Hansen* , Zhechen Yuan*, Yanjie Ze*, Tongzhou Mu*, Aravind Rajeswaran^, Hao Su^, Huazhe Xu^, Xiaolong Wang^ https://arxiv.org/abs/2212.05749	2023
MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations International Conference on Learning Representations (ICLR) Nicklas Hansen , Yixin Lin, Hao Su, Xiaolong Wang, Vikash Kumar, Aravind Rajeswaran https://arxiv.org/abs/2212.05698	2023
On the Feasibility of Cross-Task Transfer with Model-Based Reinforcement Learning International Conference on Learning Representations (ICLR) Yifan Xu*, Nicklas Hansen* , Zirui Wang, Yung-Chieh Chan, Hao Su, Zhouwen Tu https://arxiv.org/abs/2210.10763	2023

Visual Reinforcement Learning with Self-Supervised 3D Representations

IEEE Robotics and Automation Letters (RA-L)

2023

International Conference on Intelligent Robots and Systems (IROS)

Yanjie Ze*, **Nicklas Hansen***, Yinbo Chen, Mohit Jain, Xiaolong Wang

<https://arxiv.org/abs/2210.07241>

Graph Inverse Reinforcement Learning from Diverse Videos

Oral

Conference on Robot Learning (CoRL)

2022

Sateesh Kumar, Jonathan Zamora*, **Nicklas Hansen***, Rishabh Jangir, Xiaolong Wang

<https://arxiv.org/abs/2207.14299>

Temporal Difference Learning for Model Predictive Control

Spotlight

International Conference on Machine Learning (ICML)

2022

Nicklas Hansen, Xiaolong Wang*, Hao Su*

<https://arxiv.org/abs/2203.04955>

Look Closer: Bridging Egocentric and Third-Person Views with Transformers for Robotic Manipulation

IEEE Robotics and Automation Letters (RA-L)

2022

International Conference on Robotics and Automation (ICRA)

Rishabh Jangir*, **Nicklas Hansen***, Sambaran Ghosal, Mohit Jain, Xiaolong Wang

<https://arxiv.org/abs/2201.07779>

Learning Vision-Guided Quadrupedal Locomotion End-to-End with Cross-Modal Transformers

Spotlight

2022

International Conference on Learning Representations (ICLR)

Ruihan Yang*, Minghao Zhang*, **Nicklas Hansen**, Hauzhe Xu, Xiaolong Wang

<https://arxiv.org/abs/2107.03996>

Stabilizing Deep Q-Learning with ConvNets and Vision Transformers under Data Augmentation

Conference on Neural Information Processing Systems (NeurIPS)

2021

Nicklas Hansen, Hao Su, Xiaolong Wang

<https://arxiv.org/abs/2107.00644>

Generalization in Reinforcement Learning by Soft Data Augmentation

International Conference on Robotics and Automation (ICRA)

2021

Nicklas Hansen, Xiaolong Wang

<https://arxiv.org/abs/2011.13389>

Self-Supervised Policy Adaptation during Deployment

Spotlight

International Conference on Learning Representations (ICLR)

2021

Nicklas Hansen, Rishabh Jangir, Yu Sun, Guillem Alenyà, Pieter Abbeel, Alexei A. Efros,

Lerrel Pinto, Xiaolong Wang

<https://arxiv.org/abs/2007.04309>

Short Term Blood Glucose Prediction Based on Continuous Glucose Monitoring Data

IEEE Engineering in Medicine and Biology Conference (EMBC)

2020

Ali Mohebbi, Alexander R. Johansen, **Nicklas Hansen**, Peter E. Christensen, Jens M. Tarp,

Morten L. Jensen, Henrik Bengtsson, Morten Mørup

<https://arxiv.org/abs/2002.02805>

Teaching

University of California, San Diego

ECE285 Introduction to Visual Learning

· Held office hours, graded assignments.

Teaching Assistant

Spring 2024

Technical University of Denmark

Reinforcement Learning

- Special course that I co-organized w/ Prof. Ole Winther for a group of students.
- Three weeks of full-time study.

Co-organizer

Jan 2021

Technical University of Denmark

02456 Deep Learning

- Significant course material contributions, supervised 100+ students' projects on reinforcement learning.

Teaching Assistant

Fall 2019, Fall 2020

02454 Introduction to Cognitive Science

- Assisted tutorial sessions, graded assignments.

Fall 2019

Current and Former Mentees

Rishabh Jangir (MS, UCSD → ML Engineer, Skild AI)	2020 - 2022
Mohit Jain (MS, UCSD → ML Engineer, Pinterest)	2020 - 2022
Xinyue Chen (BS, NYU Shanghai → PhD, UC Berkeley)	2021 - 2022
Sateesh Kumar (MS, UCSD → PhD, UT Austin)	2021 - 2023
Jonathan Zamora-Anaya (BS, UCSD → MS, USC)	2021 - 2023
Sambaran Ghosal (MS, UCSD → ML Engineer, BrainCorp)	2021 - 2023
Zirui "Colin" Wang (BS, UCSD → PhD, UC Berkeley)	2022 - 2023
Ziyan Xiong (BS, Tsinghua University → PhD, UMass)	2022 - 2023
Yanjie Ze (BS, SJTU → PhD, Stanford)	2021 - 2023
Yunhai Feng (MS, UCSD → PhD, Cornell)	2022 - 2024
Chandramouli Rajagopalan (MS, UCSD → ML Engineer, Streamingo)	2022 - 2024
Jyothis S V (MS, NYU → RA, NYU)	2023 - 2024
Adrià Lopez (MS, ETH → PhD, ETH)	2024 - 2025
Nathaniel Thomas (MS, UCSD → ML Engineer, Assured Robot Intelligence)	2024 - 2026
Yutao Xie (MS, UCSD → PhD, UC Santa Barbara)	2024 - 2026
Jiajun Xi (MS, UCSD)	2024 - 2026
Varun Giri (MS, Georgia Tech)	2024 -
Anastasiia Pedan (MS, U. of Alberta)	2025 -

Invited Talks

Sony AI	"Improving agents via online interaction"	May 2026
Univ. Michigan	"Robots that learn by doing"	Apr 2026
UC Riverside	"AI agents should learn by doing"	Apr 2026
CMU Intent Lab	"Improving agents via online interaction"	Feb 2026
U. Washington	"Improving agents via online interaction"	Feb 2026
Johns Hopkins U.	"Improving agents via online interaction"	Feb 2026
Stanford REAL	"Improving agents via online interaction"	Feb 2026
NYU CILVR	"Improving agents via online interaction"	Feb 2026
UC Berkeley RLL	"Building Massively Multitask World Models"	Jan 2026
EWM @ NeurIPS	"Massively Multitask World Models for Continuous Control"	Dec 2025
RIWM @ ICCV	"Massively Multitask World Models for Continuous Control"	Oct 2025
Cohere Labs	"World Models on an Academic Budget"	Aug 2025
CU Hong Kong	"Learning World Models for Embodied AI"	May 2025
World Models @ ICLR	"World Models on a Budget"	Apr 2025
UIUC CSLSC	"Learning World Models for Robots"	Feb 2025
KEA Copenhagen	"Data-Driven Algorithms for Robotics"	Dec 2024
USC	"Data-Driven World Models for Robots"	Oct 2024
HuggingFace	"TD-MPC, TD-MPC2, and beyond"	Jun 2024
BeNeRL Seminar	"Data-Driven World Models for Robots"	Jun 2024

TILOS Institute	"Large Datasets and Models for Robots in the Real World"	May 2024
Univ. Michigan	"Robot Learning with (Generalist) World Models"	Jan 2024
Georgia Tech	"Building Generalist World Models"	Jan 2024
Tech. Univ. Denmark	"Data-Driven World Models at Scale: Why, What, and How?"	Dec 2023
Tsinghua IIIS	"The Next Generation of World Models"	Mar 2023
MILA/ServiceNow	"World Models with Behavioral Priors"	Feb 2023
Georgia Tech	"Towards Sample-Efficient Robot Learning with World Models"	Jan 2023
Meta AI (FAIR)	"Pretraining for Control: Current Challenges and Solutions"	Jan 2023
TU Delft	"Model-Based Reinforcement Learning: A Path Towards Generalist Agents?"	Oct 2022
Generally Intelligent	Podcast: https://generallyintelligent.com/podcast/2022-12-16-podcast-episode-25-nicklas-hansen/	Sep 2022
Intel AI	"Temporal Difference Learning for Model Predictive Control"	Apr 2022
Intel AI	"Robots that Generalize"	Aug 2021
G-Research	"Agents that Generalize and Adapt"	Feb 2021

Academic Service

Workshop organizer

2nd Workshop on World Models (ICLR)	2026
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Journal reviewer

Robotics and Autonomous Systems (RAS)	2026
IEEE Transactions on Robotics (T-RO)	2024
Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2024
Journal of Machine Learning Research (JMLR)	2023
International Journal of Computer Vision (IJCV)	2023

Conference reviewer

Conference on Neural Information Processing Systems (NeurIPS)	2022, 2023, 2024, 2025, 2026
International Conference on Machine Learning (ICML)	2022, 2023, 2025, 2026
International Conference on Learning Representations (ICLR)	2024, 2025, 2026
Conference on Computer Vision and Pattern Recognition (CVPR)	2022, 2023, 2026
European Conference on Computer Vision (ECCV)	2022, 2026
International Conference on Computer Vision (ICCV)	2023
IEEE Robotics and Automation Letters (RA-L)	2021, 2022, 2023, 2024, 2025
IEEE International Conference on Robotics & Automation (ICRA)	2023, 2024, 2026
Annual Conference on Robot Learning (CoRL)	2025
International Conference on Intelligent Robots and Systems (IROS)	2023, 2025, 2026
Association for the Advancement of Artificial Intelligence (AAAI)	2021

Workshop reviewer

Workshop on X-Embodiment Robot Learning @ CoRL	2024
Foundation Models for Decision-Making, Workshop @ NeurIPS	2023
Self-Supervised Learning - Theory and Practice, Workshop @ NeurIPS	2023
Learning Dexterous Manipulation, Workshop @ RSS	2023
Structural and Compositional Learning on 3D data, Workshop @ CVPR	2023
Self-Supervised Learning - Theory and Practice, Workshop @ NeurIPS	2022
Generalizable Policy Learning in the Physical World, Workshop @ ICLR	2022

Workshop Presentations

Generalizable Robotic Insertion with World Models	Poster
Workshop on Out-of-Distribution Generalization in Robotics @ RSS	2025

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning

Poster

Workshop on Out-of-Distribution Generalization in Robotics @ RSS

2025

Musculoskeletal Control with TD-MPC2

Poster

MyoSuite Symposium @ NeurIPS (2nd place Locomotion track + Physiological Award)

2024

TD-MPC2: Scalable, Robust World Models for Continuous Control

Poster

Foundation Models for Decision-Making @ NeurIPS

2023

Robot Learning @ NeurIPS

2023

Pre-Training Robot Learning @ CoRL

2023

On Pre-Training for Visuo-Motor Control: Revisiting a Learning-from-Scratch Baseline

Poster

Pre-Training Robot Learning @ CoRL

2022

On the Feasibility of Cross-Task Transfer with Model-Based Reinforcement Learning

Poster

Pre-Training Robot Learning @ CoRL

2022

Foundation Models for Decision Making @ NeurIPS

2022

Deep RL Workshop @ NeurIPS

2022

MoDem: Accelerating Visual Model-Based Reinforcement Learning with Demonstrations

Poster

Pre-Training Robot Learning @ CoRL

2022

Deep RL Workshop @ NeurIPS

2022

Look Closer: Bridging Egocentric and Third-Person Views with Transformers for Robotic Manipulation

Poster

Workshop on Deployable Decision Making in Embodied Systems @ NeurIPS

2021

Deep RL Workshop @ NeurIPS

2021

Learning Vision-Guided Quadrupedal Locomotion End-to-End with Cross-Modal Transformers

Poster

Deep RL Workshop @ NeurIPS

2021

Visual Learning and Reasoning for Robotics Workshop @ RSS

2021

Stabilizing Deep Q-Learning with ConvNets and Vision Transformers under Data Augmentation

Poster

Unsupervised RL Workshop @ ICML

2021

Visual Learning and Reasoning for Robotics Workshop @ RSS

2021

Self-Supervised Policy Adaptation During Deployment

Poster

Microsoft Research RL Day

2021

Deep RL Workshop @ NeurIPS

2020

Workshop on Robot Learning @ NeurIPS

2020

Work Experience

NVIDIA Research

Seattle, WA, USA

Research Intern

June 2024 - Feb 2025

· Robot learning. Mentored by Yashraj Narang and Dieter Fox.

Meta AI (FAIR)

Menlo Park, CA, USA

Research Scientist Intern

June 2022 - Dec 2022

· Model-based reinforcement learning. Mentored by Aravind Rajeswaran.

raffle.ai

Copenhagen, Denmark

Machine Learning Intern

Summer 2019

· I built and open-sourced a cross-domain text-to-SQL model in PyTorch.

Retune DSP

Student Assistant

· I helped a team of engineers build and maintain deep learning pipelines for embedded voice control.

Kongens Lyngby, Denmark

Feb 2019 - Dec 2019

Nordic Transition

Student Software Developer

· I developed and maintained a data management and analysis platform for the HR industry.

Gentofte, Denmark

July 2016 - Dec 2019

Awards and Scholarships

2026	UC Riverside Future Academics & Mentorship Excellence (FAME) Program	Award
	· A 2-day workshop for future faculty applicants (8 recipients nation-wide).	
2024	NVIDIA Graduate Fellowship	Fellowship
	· An award of \$60,000 to cover stipend and tuition for 1 year (10 recipients worldwide).	
2023	NeurIPS Top Reviewer	Award
	· Awarded free conference registration as a top peer reviewer for the conference.	
2021	Robotics Summer School Scholarship	Scholarship
	· A scholarship to participate in a two-week summer program in Denmark.	
2020	Spar Nord Fond Scholarship	Scholarship
	· A scholarship to study a semester at UC Berkeley (5 recipients nation-wide).	
2020	UC Berkeley's SCET Collider Cup Finalist	Award
	· Biannual startup competition. Best student project from each class is nominated.	
2017	Otto Mønsted Fonds Legat	Scholarship
	· A grant for students with a GPA ≥ 8.0 who wish to study a semester abroad.	

Volunteering

2026	UC San Diego, CSE PhD Visit Days Organizer	Organizing
	· Co-organizing visit days for prospective PhD students in the CSE department.	
2025	NeurIPS, Volunteer	Organizing
	· Volunteer work for NeurIPS 2025 in San Diego.	
2024	Queer in AI, Graduate Application Financial Assistance	Organizing
	· Reviewed applications for microgrants associated with graduate applications.	
2024	UC San Diego, Graduate Women in Computing Mentor	Mentorship
	· Mentor for first-year PhD students that identify with a gender minority.	
2023	UC San Diego, GradAMP Mentor (PhD Applications)	Mentorship
2022	· Supported prospective students through weekly mentor-mentee meetings in Fall.	

Misc. Open-Source Projects

TD-MPC2 Official Implementation (★820)	2024
· Public code release for "TD-MPC2: Scalable, Robust World Models for Continuous Control". https://github.com/nicklashansen/tdmpc2	
TD-MPC Official Implementation (★513)	2022
· Public code release for "Temporal Difference Learning for Model Predictive Control". https://github.com/nicklashansen/tdmpc	
DMControl Generalization Benchmark (★189)	2020
· Benchmark for generalization in continuous control from pixels. https://github.com/nicklashansen/dmcontrol-generalization-benchmark	